

Agent-Based Defense Against Label-Flipping Poisoning Attacks in EEG Emotion Classification



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Model I vs Model II vs Model III

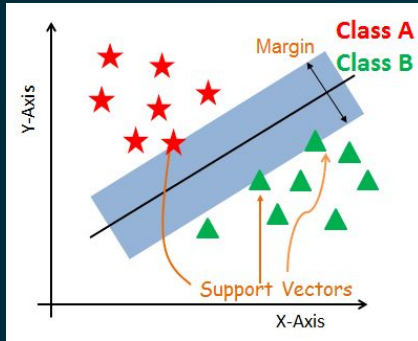


01

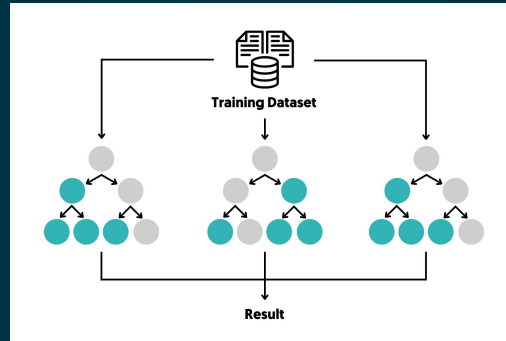
BASELINE MODEL (MODEL I)

Baseline Model - Model I

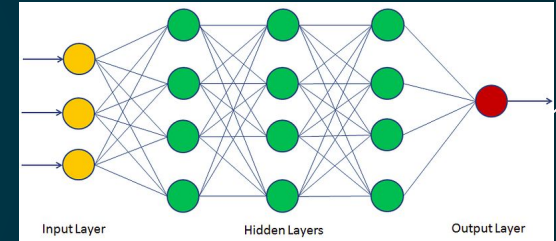
Support Vector Machine



Random Forest



Multi-Layer Perceptron

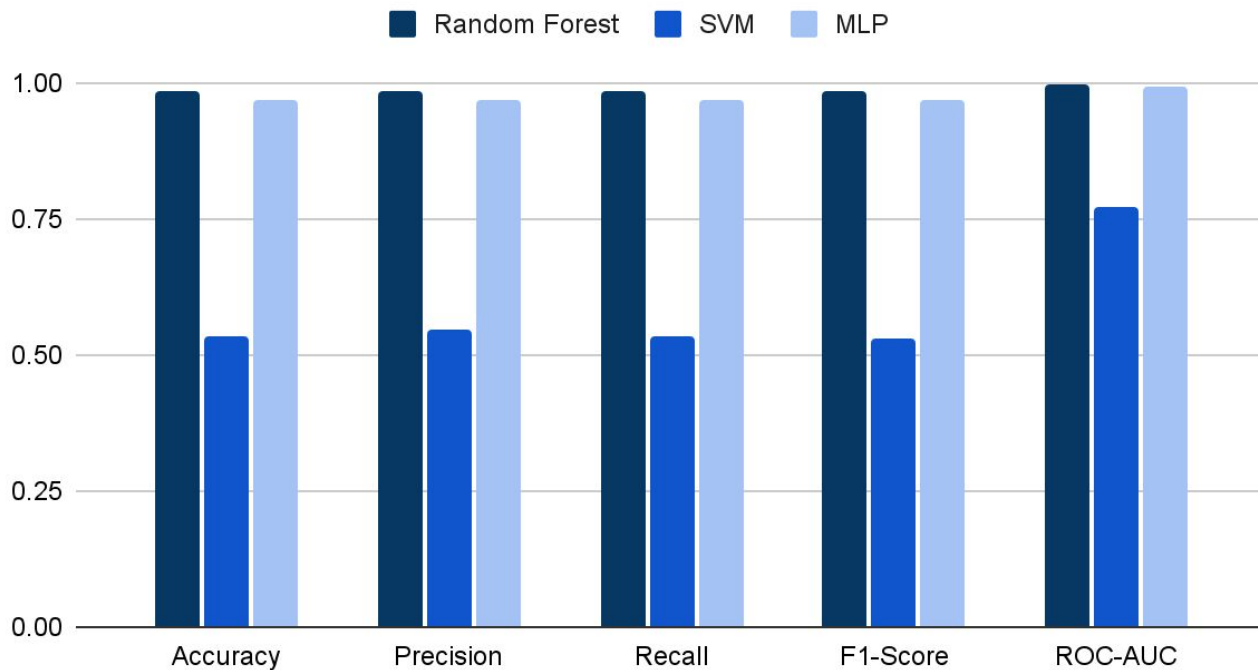


Baseline Model - Model I

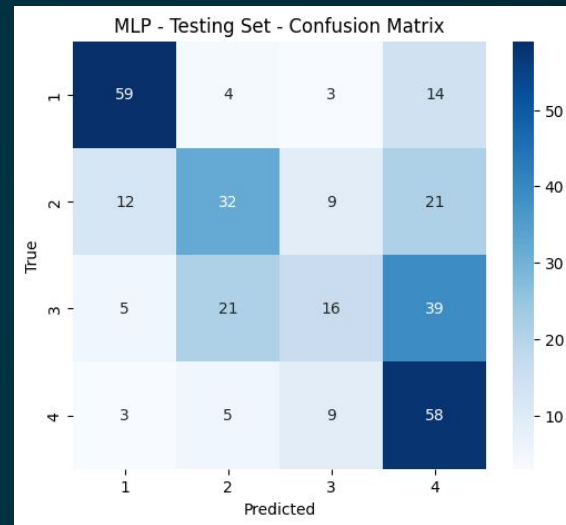
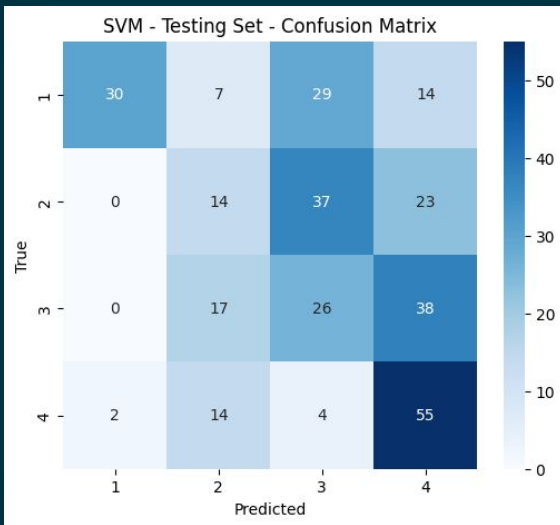
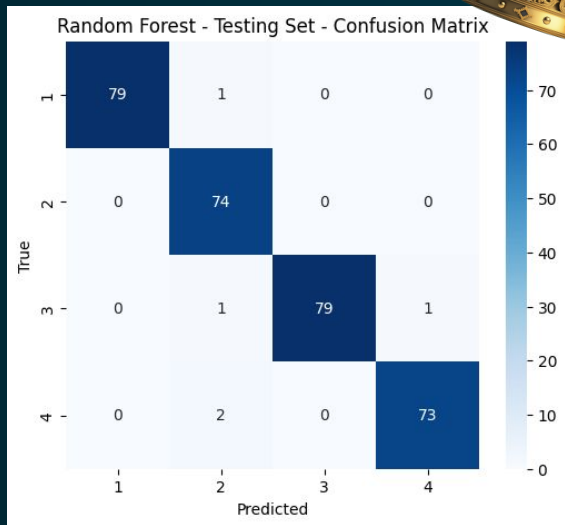
Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.9833	0.9842	0.9833	0.9833	0.9988
SVM	0.5356	0.5465	0.5356	0.5319	0.7736
MLP	0.9665	0.9677	0.9665	0.9667	0.9930

Baseline Model - Model I

Comparison of Baseline Models



Baseline Model - Model I





02

DATA POISONING ATTACK (MODEL II)

Data Poisoning Attack - Model II

Goal: stress-test baseline RF against training-time label corruption

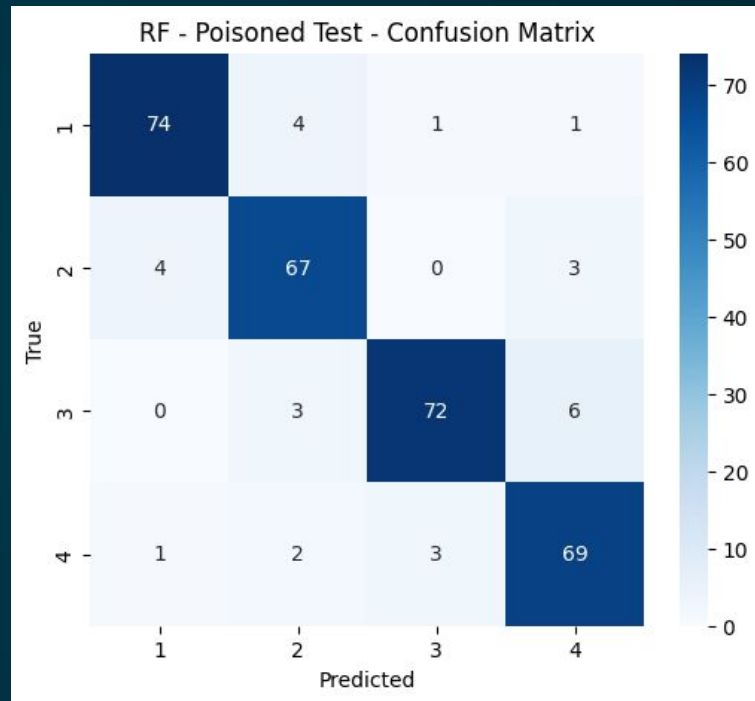
Attack Type: Label Flipping

Poisoning Rate: 15% (186 mislabeled training samples)

```
def poison_labels(y, poison_rate=0.15, random_state=42):  
  
    rng = np.random.RandomState(random_state) #for reproducible randomness  
  
    y = np.array(y)  
    y_poisoned = y.copy()  
  
    n_samples = len(y)  
    n_poison = int(np.floor(poison_rate * n_samples)) #nr of labels to flip  
  
    #randomly choose which samples to poison  
    all_indices = np.arange(n_samples)  
    poison_idx = rng.choice(all_indices, size=n_poison, replace=False)  
  
    classes = np.unique(y) #all possible emotion labels  
  
    #flipping labels  
    for idx in poison_idx:  
        current_label = y[idx]  
        #pick a diff. label  
        possible_labels = [c for c in classes if c != current_label]  
        new_label = rng.choice(possible_labels)  
        y_poisoned[idx] = new_label  
  
    return y_poisoned, poison_idx
```

Metrics Evaluation - Model II

- Accuracy: 0.9097
- Precision: 0.9110
- Recall: 0.9097
- F1 Score: 0.9099
- ROC-AUC: 0.9893



A collection of isometric icons in shades of teal and light blue. The icons include several cloud shapes, a shield with a checkmark, a padlock, and two speech bubbles. These icons are arranged around the central text.

03

AGENT-BASED DEFENSE PIPELINE (MODEL III)



Label Curator Agent

01

Detects Mislabels

Identifies samples whose assigned label strongly disagrees with the baseline model's prediction.

02

Correct Labels

Relabels suspicious points using the clean Model 1 as a reference.

03

Recovers Label Integrity

Reduces the number of poisoned or inconsistent labels introduced by the attack.

04

First Step of the Defense Pipeline

Creates a more trustworthy dataset that is passed to the next agent.



Centroid-based Curator Agent

01

Computes Centroids

Calculates the average feature vector for each emotion class.

02

Measure Distances

Measures the distance of each point to its class centroid to detect structural anomalies.

03

Removes Outliers

Eliminates the top fraction of points with the largest distances, targeting feature-space inconsistencies.

04

Cleaner Feature Space

Produces a more coherent representation of each class before validation.



Validator Agent

01

Evaluates Datasets

Trains a lightweight Logistic Regression model on both the Label-Only and Label+Centroid cleaned datasets.

02

Compares Performance

Measures which dataset gives higher validation accuracy label-check-only or the combined curators.

03

Selects Best Data

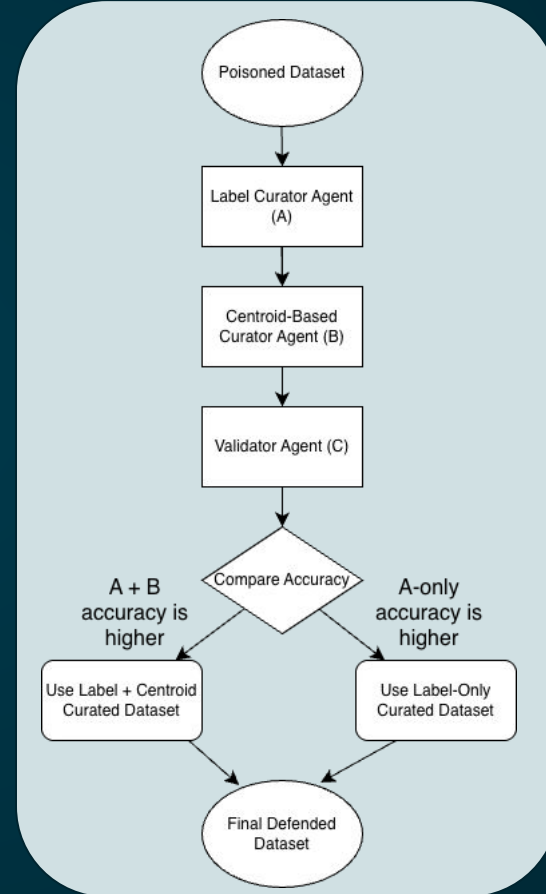
Chooses the dataset that generalizes better.

04

Final Output

Delivers the dataset used to train the defended Random Forest (Model III).

Overall Defense Flow



A collection of isometric icons in shades of teal and light blue. The icons include several cloud shapes of different sizes, a large shield with a checkmark inside, a padlock, and two speech bubbles. These icons are arranged in a cluster in the upper center of the slide.

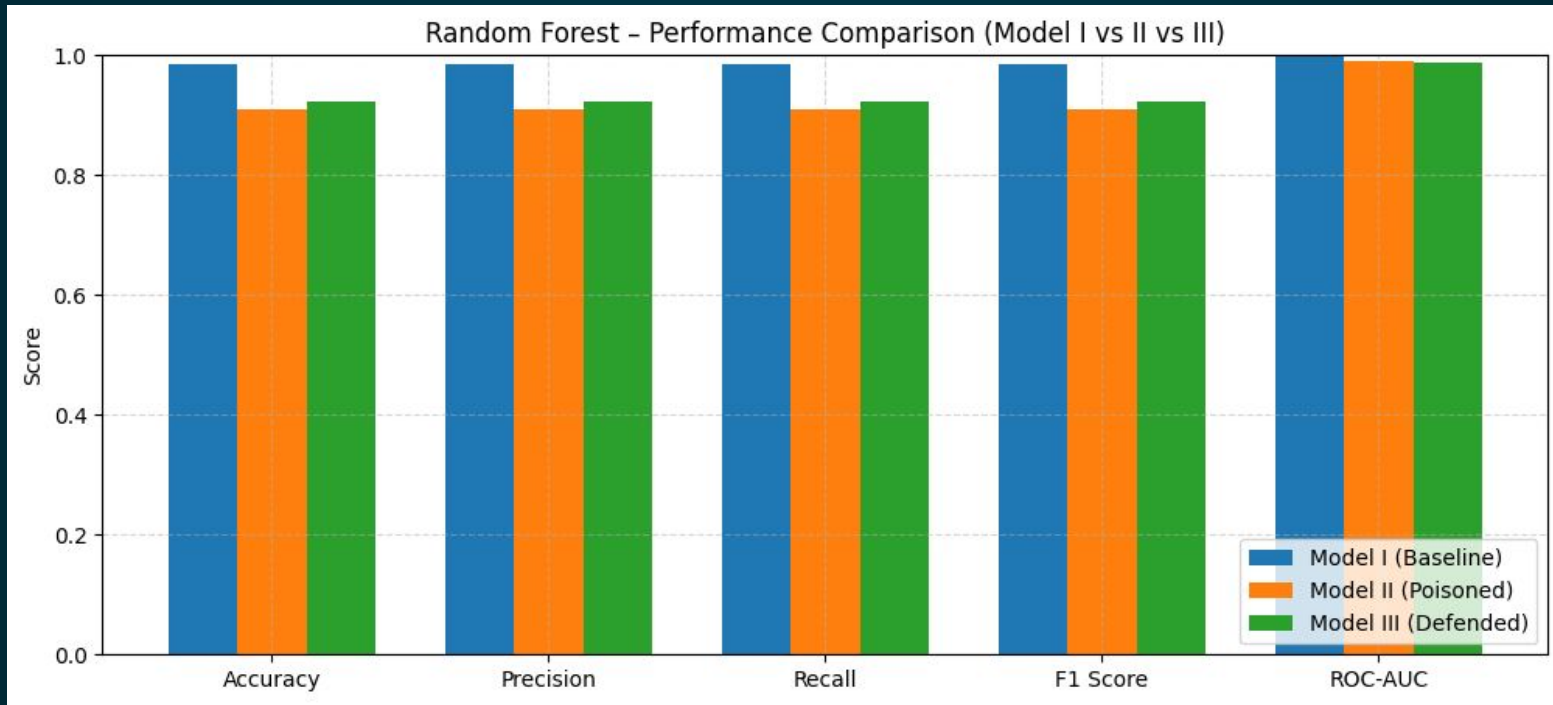
04

PERFORMANCE COMPARISON

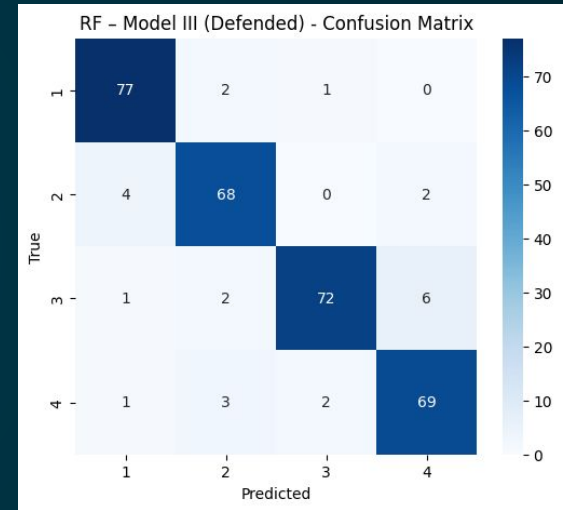
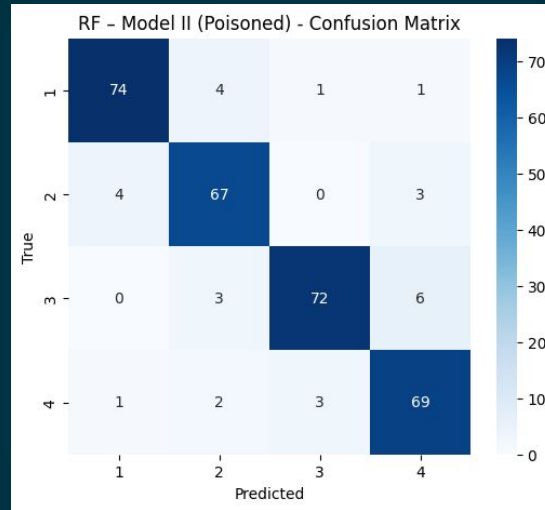
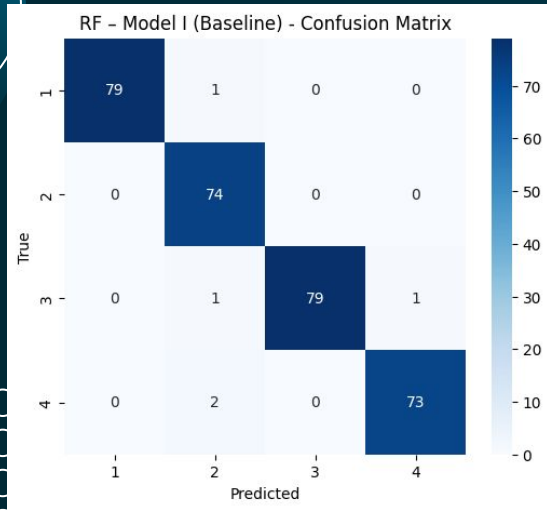
Evaluation Metrics of All Models

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Model I	0.9839	0.9845	0.9839	0.9840	0.9993
Model II	0.9097	0.9110	0.9097	0.9099	0.9893
Model III	0.9226	0.9235	0.9226	0.9225	0.9882

Side-by-Side Performance Comparison



Confusion Matrix Comparison





THANK YOU!